README for "Data from Glacier Model Intercomparison Project Phase 3 (GlacierMIP3)" doi: 10.5281/zenodo.14045268

This document provides data documentation for the Glacier Model Intercomparison Project 3 (GlacierMIP3), which focuses on global glacier mass change equilibration experiments.

GlacierMIP is a framework for a coordinated intercomparison of global-scale glacier mass change models to foster model improvements and reduce uncertainties in global glacier projections. It is running as a 'Targeted Activity' under the auspices of the Climate and Cryosphere Project CliC, a core project of the World Climate Research Programme (WCRP).

For detailed information about the GlacierMIP3 experimental design, please refer to the GlacierMIP3 protocol.

If you are only interested in the data from Supplementary Table 1 or 3, you can directly go to the corresponding CSV files (`table_S1a.csv`, `table_S1b.csv`, `table_S3.csv`).

The dataset includes the regional glacier volume and area projections as submitted by the glacier modelling groups, encapsulated in the main file `GMIP3_reg_glacier_model_data/glacierMIP3_Feb12_2024_models_all_rgi_regions_sum.nc`. Additionally, it features post-processed and aggregated data derived from GlacierMIP3, or in combination with other studies, which is used for the analyses and visualisations presented in the manuscript.

> Zekollari*, H., Schuster*, L., Maussion, F., Hock, R., Marzeion, B., Rounce, D. R., Compagno, L., Fujita, K., Huss, M., James, M., Kraaijenbrink, P. D. A., Lipscomb, W. H., Minallah, S., Oberrauch, M., Van Tricht, L., Champollion, N., Edwards, T., Farinotti, D., Immerzeel, W., Leguy, G., Sakai, A. (under review). Glacier preservation doubled by limiting warming to 1.5°C (https://doi.org/10.31223/X51T5W), 2024.
> *These authors contributed equally

When using this dataset, please cite both the Zenodo dataset (doi: 10.5281/zenodo.14045268) and the submitted study above. Note that we are currently working on another potential study to analyse the glacier model differences more thoroughly.

To assist potential data users, we have included a jupyter notebook (`gmip3_data_example_use_cases.ipynb`) that guides you through some simple use cases. This notebook can be directly run when clicking on this link. Please note that you will need to log in to your Google account and, if you haven't already done so, install Google Colaboratory. The data will then be automatically downloaded to your account.

We may adapt the data structure and improve the documentation during the review phase. If you have any questions or suggestions, please contact us.

Please note that the data described is only available through the published Zenodo dataset. The code used to generate the postprocessed data and to conduct the analyses for the manuscript mentioned above is available at https://github.com/GlacierMIP/GlacierMIP3. We have also retained this README_data in the GitHub repository for reference.

Below, you will find the documentation for the different individual datasets.

GMIP3_reg_glacier_model_data/

Here, the regionally aggregated glacier model projections (volume and area) are given in different postprocessing steps (explained in Methods and Supplementary Data Table 2 of Zekollari, Schuster et al., submitted).

All files within the `GMIP3_reg_glacier_model_data` folder are netCDF files with data for volume in m³ (`volume_m3`) and area in m² (`area_m2`) which have the following dimensions:

Variable	Dimension	Description
`model_author`	8 (10)	One of the eight model authors from the list: ['PyGEM-OGGM_v13', 'GloGEMflow', 'GloGEMflow3D', 'OGGM_v16', 'GLIMB', 'Kraaijenbrink', 'GO', 'CISM2']. In the non-final postprocessed files, we included two OGGM variants which we might only use for a more technical model intercomparison study (more in `gmip3_data_example_use_cases.ipynb`)
`simulation_year`	5001	Simulation year from 0 to 5000 (estimates may be NaNs after 2000)
`gcm`	5	ISIMIP3b climate model: ['gfdl-esm4', 'ipsl-cm6a-lr', 'mpi-esm1-2-hr', 'mri-esm2-0', 'ukesm1-0-ll'] (corresponding warming level available in climate_input_data/).

`period_scenario`	16	One of the 16 climate scenarios (combining time periods and scenarios): ['1851-1870_hist', '1901-1920_hist', '1951-1970_hist', '1995-2014_hist', '2021-2040_ssp126', '2021-2040_ssp370', '2021-2040_ssp585', '2041-2060_ssp126', '2041-2060_ssp370', '2041-2060_ssp585', '2061-2080_ssp126', '2061-2080_ssp370', '2061-2080_ssp585', '2081-2100_ssp126', '2081-2100_ssp370', '2081-2100_ssp585']
`rgi_reg`	19	One of the 19 RGI6 regions, from '01' to '19'.

The following different regional glacier model projections are available (ordered from the least to the most postprocessed):

`GMIP3_reg_glacier_model_data/glacierMIP3_Feb12_2024_models_all_rgi_regions_sum.nc`

- unprocessed dataset with the "raw" regional files submitted by the glacier model groups. Not used directly in any of the community estimate manuscript analyses, but of most interest for model comparison analyses
- Important additional information for OGGM: Glaciers that could not be simulated ("failing glaciers" mainly because running out of the domain) were replaced by mean estimates (volume/area) from glaciers with a similar area ("filling approach"). However, as a filling but no direct upscaling was applied, regional initial volumes might slightly differ from the regional volume estimates of Farinotti et al. 2019. Substantial differences are only for climate scenarios near preindustrial temperatures in specific regions (Iceland (RGI06), and partially Alaska (RGI01) and New Zealand (RGI18)).
- Important additional information for PyGEM-OGGM_v13: Failing glaciers were filled by scaling their initial volume or area by the normalized regional mean values of all successful glaciers. Inversion performed for every climate dataset, so initial volume may vary by simulation for the same glacier and thus also varies within one RGI region.
- > in the first manuscript, this file was only used to create the next dataset.

`GMIP3_reg_glacier_model_data/glacierMIP3_Feb12_2024_models_all_rgi_regions_sum_scaled.nc`

- same as the previous file but volume scaled to match regional Farinotti et al. (2019) multi-model glacier volume estimate at the beginning, area scaled to match the RGI6.0 area at the beginning. Scaling is done individually for each climate scenario and model time series (experiment). For most experiments, scaling was not necessary as models aimed to match regional volume/area and upscaled already internally

- > only used to estimate the number of years until steady state is reached (in `A_community_estimate_paper_analysis/2b_find_equilibrium_steady_state_yr.ipynb` for Supplementary Fig. S8) and used to create the next dataset

`GMIP3_reg_glacier_model_data/glacierMIP3_Feb12_2024_models_all_rgi_regions_sum_scaled_extended_repeat_last_101yrs.nc`

- same as the previous file but where necessary, the experiments are extended from the year 2000 to the year 5000
- > used in some notebooks for the steady-state estimates (note that the steady-state estimates are the same as in the below-shifted dataset)

`GMIP3_reg_glacier_model_data/all_shifted_glacierMIP3_Feb12_2024_models_all_rgi_regions_sum_scaled_extended_repeat_last_101yrs_via_5yravg.nc`

- same as the previous file, but the timeseries were shifted so that the initial state of the new timeseries match better the glacier mass estimate of the year 2020
- `area_m2` removed here, as not analysed in the first GlacierMIP3 study
- additional coordinate:
    - `year_after_2020`: new coordinate which describes the "simulation year" from the 2020 state onwards - unit: year
- here `simulation_year` is the initial simulation year as given by the unshifted version (and dimension changed to `year_after_2020`, see below)
- additional variables:
    - `volume_rel_2020_%`: scaled, extended and then shifted glacier volume/mass relative to 2020 - unit: %
    - `yrs_w_most_similar_state_to_2020`: year with a glacier volume/mass similar to the estimated state in 2020 - unit: year
    - `temp_ch_ipcc`: Global warming above pre-industrial (1850-1900) for that experiment, using the IPCC AR6 definition (i.e. assume +0.69°C between 1850 and 1986-2005 - unit: °C
- > used for the LOWESS fits later, but also to estimate the response timescales, and also directly for e.g. Fig. 1,2; Supplementary Fig. S5, S7

climate_input_data/

Warming above pre-industrial for the different experiments (either on a global mean average or regional).

Column Name	Description

`gcm`	ISIMIP3b climate model
`period_scenario`	One of the 16 climate scenarios (combining time periods and scenarios, e.g. `2041-2060_ssp126`)
`region`	Aggregated region over which the warming was estimated. Global mean average (`global`),
	global glacier-area weighted mean (`global_glacier`), regional glacier-area weighted mean (e.g. `RGI11_glacier`)
	`temp_ch_ipcc_ar6_isimip3b` does not have this column as all entries are `global`
`temp_ch_ipcc`	Global warming above pre-industrial (1850-1900) using the IPCC AR6 definition (i.e. assume +0.69°C between 1850 and 1986-2005 - unit: °C

The following different CSV files are available:

`climate_input_data/temp_ch_ipcc_ar6_isimip3b.csv`: only the global mean average warming

- > used in most figures and analysis

`climate_input_data/temp_ch_ipcc_ar6_isimip3b_glacier_regionally.csv`: all regions

- Attention: Here we assume regionally equally +0.69°C of warming between 1850 and 1986 to 2005. For the warming ratio computation, shown in e.g. Fig. 2 or Fig. 4, we subtract the 0.69°C again to instead just show the regional warming ratio to the actual regional 1986 to 2005 estimates.
- > used e.g. for Supplementary Fig. S2, and to create `3_shift_summary_region_characteristics.csv`

LOWESS fits

The relative glacier mass changes were LOWESS fitted with the respective temperature changes of the 80 climate scenarios. The LOWESS fit results, here given in steps of 0.05°C, are given in the `lowess_fit_rel_2020*.csv` files for different variants.

All `lowess_fit_rel_2020*.csv` files have the following columns:

Variable	Description
`temp_ch`	Warming level above pre-industrial in °C.

`0.5`, `0.17`, `0.83` (and others)	Quantiles of fitted remaining glacier mass in % relative to 2020 (21- or 101-year rolling average) for the respective region and warming level.
`region`	The 19 RGI6 regions ('01' to '19') or all regions together ('All'), which corresponds to the global estimates.
`year`	100, 500, or 5000 - Year of analysis.
`frac`	Chosen frac parameter (hyperparameter of how many data points are applied for the local regression, see MOEPY package documentation).
	Note: frac is only valid for the median values in case of `region=='All'`.
`it`	Number of iterations for the lowess fit (see MOEPY package).
`N`	Repetitions (see MOEPY package).

The following different LOWESS fitted data files are available:

`lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024.csv`

- fit uses relative remaining mass at steady-state (101-year rolling average of the last simulation years, either after 5000 or 2000 years) together with global mean warming above preindustrial
- year is here 5000 (although for some regions simulations were only done for 2000 years)
- > used for Fig. 1-4, Supplementary Table S1, and several supplementary figures

`lowess_fit_rel_2020_21yr_avg_after{YEAR}yr_Feb12_2024.csv` with YEAR=100 or YEAR=500

- similar to `lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024.csv`, but for the estimates after 100 or 500 simulation years. Quantiles were LOWESS-fitted with 21-year centered rolling average glacier mass estimates
- year is here 100 or 500
- > used for Fig. 3, Supplementary Fig. S6

`lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024_only_global_models.csv`

- similar to `lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024.csv`, but using for the LOWESS fit only globally available glacier models
- year is here 5000

> used for e.g. Supplementary Fig. S13

`lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024_rel_regional_glacier_temp_ch.csv`

- similar to `lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024.csv`, but using instead regional glacier-area weighted warming
- March 2025: For the regional warming lowess fits, it does not makes sense to aggregate to global uncertainties as we have to add them up via the composites. The reason is the different regional warming. Therefore, global uncertainties were set to "NaN" values.

> used for e.g. Supplementary Fig. S4

`lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024_per_glac_model.csv`

- similar to `lowess_fit_rel_2020_101yr_avg_steady_state_Feb12_2024.csv`, but doing a LOWESS fit for every glacier model individually.
- Note that the fit does not perform well for some glacier models below 1.5°C, and should thus not be used in that range.
- > used to estimate the spread/uncertainties in the committed ice loss sensitivity (only in range 1.5 to 3.0°C, e.g. shown in Supplementary Data Table S1, Supplementary Fig. S5)

other files

response timescale dataset CSV-file:
`resp_time_shifted_for_deltaT_rgi_reg_roll_volume_21yravg.csv` with the following columns

Column Name	Description
`rgi_reg`	RGI region (e.g. 'RGI11') or global ('All')
`model_author`	Model name (e.g. 'GLIMB')
`temp_ch_ipcc`	Global warming above pre-industrial for that experiment - unit: °C
`gcm_period_scenario`	One of the 80 experiments, e.g. in the format: 'ipsl-cm6a-lr_2081-2100_ssp370'
`min_perc_change`	Minimum shrinkage/growing limit to estimate a "response timescale", here everywhere 25% to reduce noise from interannual variability - unit: %
`resp_time_-50%`	Response timescale for -50% of the total changes to occur

	(here only if changes are losses) - unit: years
`resp_time_-80%`	Response timescale for -80% of the total changes to occur (here only if changes are losses) - unit: years
`resp_time_-90%`	Response timescale for -90% of the total changes to occur (here only if changes are losses) - unit: years
`diff_resp_time_-50%`	Difference of that model's response timescale to the median glacier model response time scale) - unit: years
`diff_resp_time_-80%`	Difference of that model's response timescale to the median glacier model response time scale) - unit: years
`diff_resp_time_-90%`	Difference of that model's response timescale to the median glacier model response time scale) - unit: years

> This dataset is used, e.g. for the following figures: Supplementary Fig. S11, S12, S15

aggregated regional characteristics summary csv-file: `3_shift_summary_region_characteristics.csv` with the following columns

Column Name	Description
`rgi_reg`, `region`	Regions e.g. ('11'), all regions together described as `All` in `rgi_reg` and as `global` in `region`
Climate Indices (data from ISIMIP3a (GSWP3-W5E5)):	
`temp_ch_avg_2000-2019_vs_1901-1920`	ΔTemp 2000-2019 - 1901-1920 (reg-aw) - unit: °C
`temp_avg_2000-2019`	Temp 2000-2019 (reg-aw) - unit: °C
`prcp_avg_2000-2019`	Prcp 2000-2019 (reg-aw) - unit: kg m-2 s-1
`continentality_index_avg_2000-2019`	Continentality index 2000-2019 (temp difference between coldest/warmest month same year, reg-aw) - unit: °C
`median_reg_vs_glob_ch`	Ratio reg vs global ΔTemp to pre-industrial (median, reg-aw) - no unit

`median_reg_vs_glob_temp_ch_1.5_3.0`	Same as `median_reg_vs_glob_ch` but only with experiments between 1.5°C and 3.0°C - no unit
`median_reg_vs_glob_ch_ref_1986-2005`	Ratio reg vs global ΔTemp to 1986-2005 (median, reg-aw) - no unit
`median_reg_vs_glob_temp_ch_1.5_3.0_ref_1986-2005`	Same as `median_reg_vs_glob_ch_ref_1986-2005` but only with experiments between 1.5°C and 3.0°C - no unit
`slope_fit_reg_vs_glob_ch`	Fitted linear fit slope regional vs global temp change (aw) - no unit
Glacier topography (data from RGI6):	
`slope_weighted_area_avg`	Glacier surface slope (reg-aw) - unit: °
`lat_absolute_weighted_area_avg`	Latitude (absolute, reg-aw) - unit: °
`lat_weighted_area_avg`	Latitude (reg-aw) - unit: °
`marine_term_ratio_hundredlargest_glac`	Ratio marine-terminating (100 largest glaciers) - no unit
`mean_vol_ten_largest_glac`	Mean volume (10 largest glaciers by area) - unit: km3
`elev_diff_area_weighted`	Elevation range (reg-aw) - unit: m
`ice_cap_ratio_hundredlargest_glac`	Ratio ice caps (100 largest glaciers) - no unit
`max_elev_area_weighted`	Maximum elevation (reg-aw) - unit: m
`min_elev_area_weighted`	Minimum elevation (reg-aw) - unit: m
Observed glacier changes and states in the past (data from Hugonnet et al. 2021 & Farinotti et al. 2019):	
`geodetic_obs_area_weighted`	Observed geodetic MB (2000-2019, reg) from Hugonnet et al.

	(2021) - unit: m w.e. year-1
`dvoldt_m3_hugonnet`	Observed ΔVolumeΔt (2000-2019, reg) from Hugonnet et al. (2021) - unit: m3 per year
`20yr_regional_dvol_dt_2000_2019_vs_2000_vol_%`	Observed ΔMassΔt (2000-2019 relative to 2000 Mass, reg) - unit: %
`regional_volume_m3_itmix`	Glacier volume as given by Farinotti et al. (2019) (reg) - unit: m3
`regional_volume_m3_2020_via_5yravg`	Glacier volume in 2020 (est. from Farinotti et al. 2019 & Hugonnet et al., 2021) (reg) - unit: m3
`regional_volume_m3_itmix_vs_2020`	Ratio of glacier volume at RGI6 inventory date vs volume in 2020 (est. from Farinotti et al. 2019 & Hugonnet et al., 2021) (reg) - no unit
`rgi_year_weighted_median`	RGI year (reg-aw) - unit: year
Glacier simulation change estimates (data from GlacierMIP3):	
`resp_time_-50%_1_5_deg`	Response timescale (~1.5°C, 50%, reg) - unit: years
`resp_time_-50%_3_0_deg`	Response timescale (~3.0°C, 50%, reg) - unit: years
`resp_time_-80%_1_5_deg`	Response timescale (~1.5°C, 80%, reg) - unit: years
`resp_time_-80%_3_0_deg`	Response timescale (~3.0°C, 80%, reg) - unit: years
`resp_time_-50%_1_5_deg_only_global_models`	Same as above but only considering glacier models with global simulations
`resp_time_-50%_3_0_deg_only_global_models`	Same as above
`resp_time_-80%_1_5_deg_only_global_models`	Same as above

`resp_time_-80%_3_0_deg_only_global_models`	Same as above
`ice_loss_1.2°C_%_rel_2020`	Committed glacier mass loss at ΔTemp=+1.2°C (rel. to 2020) - unit: %
`ice_loss_slope_1.5_to_3.0_per_tenth_degC_rel_2020`	Sensitivity of committed glacier mass loss (ΔTemp=1.5 to 3.0°C) - unit : % per 0.1°C

Abbreviations: “10/100 largest” refers to the 10/100 glaciers with the largest initial glacier mass at inventory date according to the estimate by Farinotti et al. 2021. ”reg-aw” refers to regionally glacier-area weighted, “reg” refers to regional, “avg” refers to average, “Temp” refers to temperature, “Prcp” refers to precipitation.

> used for Supplementary Data Table S1 and S3, and for Fig. 4, Supplementary Fig. S3, S9, and many other figures

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Some additional notes

For a few supplementary figures, we also used data from other studies. Sometimes we got the data via personal communication or by aggregating raw files from published datasets. These files are shortly described in `data_from_others/README_data_from_others.md`. However, they are only available via https://cluster.klima.uni-bremen.de/~lschuster/GlacierMIP3/data/data_from_others.

The per-glacier model glacier volume and area estimates are currecntly only available for the models `PyGEM-OGGM_v13` and `OGGM_v16`. If you have are interested in the glacier-specific files, please contact us.